

■ Signature

`Signature[list]` gives the signature of the permutation needed to place the elements of *list* in canonical order. by *list*.

Examples: `Signature[{a,b,c}]` $\longrightarrow$ 1; `Signature[{a,c,b}]` $\longrightarrow$ -1. ■ The signature of the permutation is $(-1)^n$, where n is the number of transpositions of pairs of elements that must be composed to build up the permutation. ■ If any two elements of *list* are the same, `Signature[list]` gives 0. ■ See pages 106 and 460. ■ See also: `Sort`.